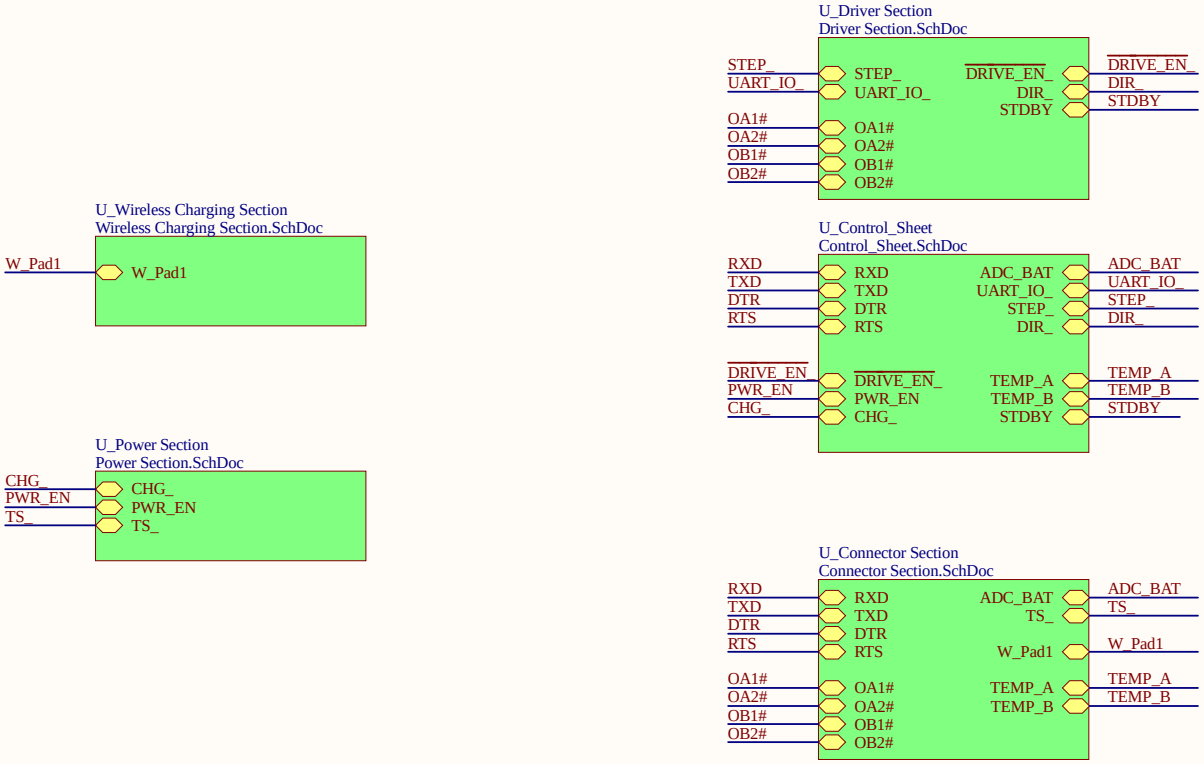
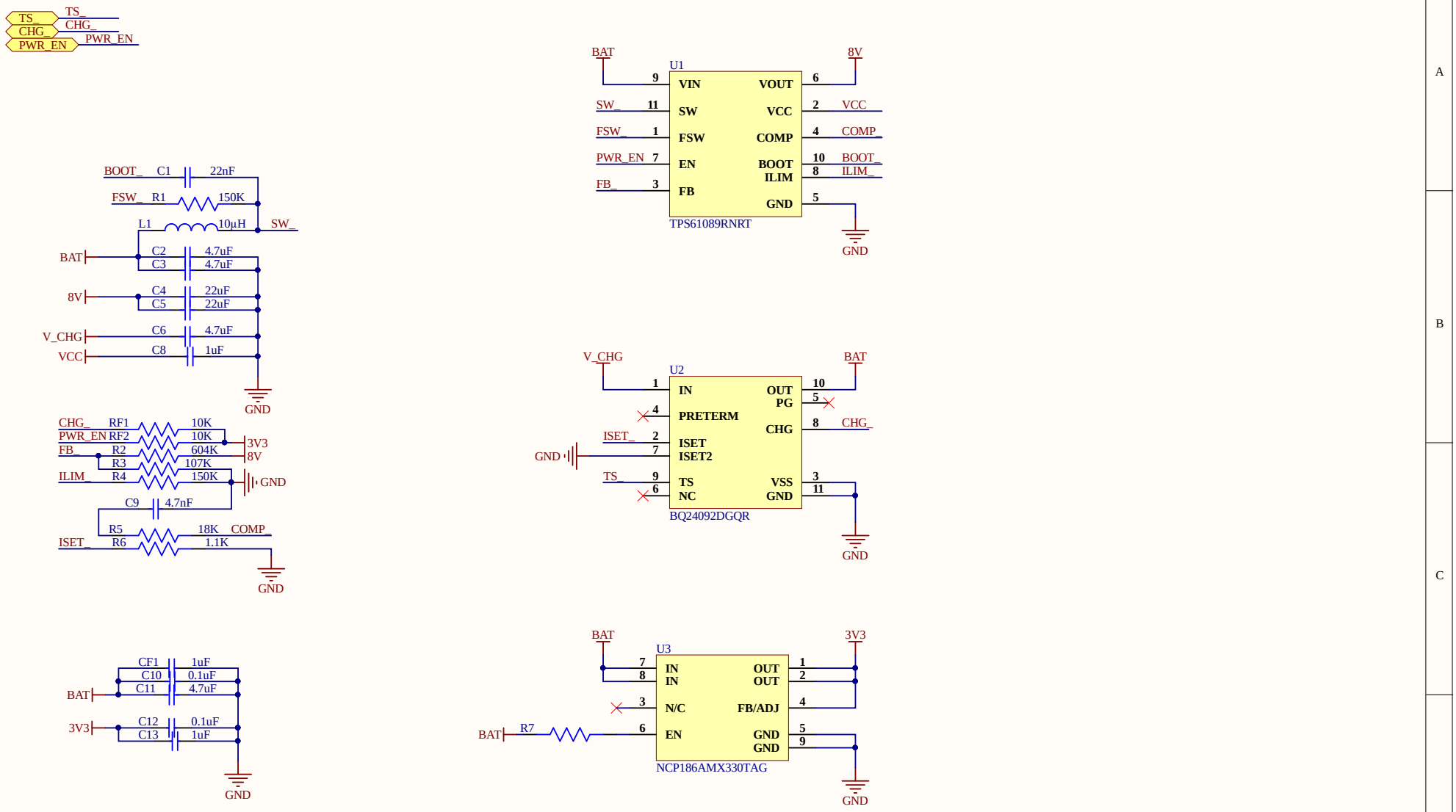
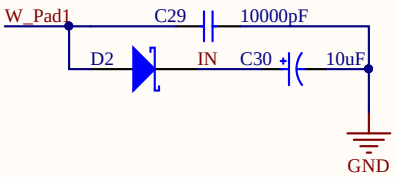
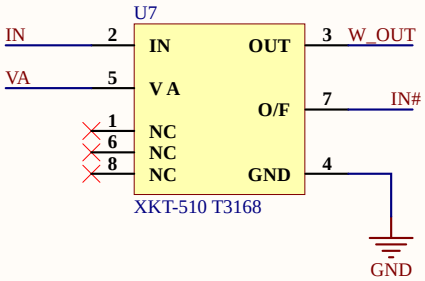

U_Wireless Charging Section
Wireless Charging Section.SchDoc
W_Pad1
W_Pad1
U_Power Section
Power Section.SchDoc
CHG_
PWR_EN
TS_
CHG_
PWR_EN
TS_
U_Driver Section
Driver Section.SchDoc
STEP_
UART_IO_
OA1#
OA2#
OB1#
OB2#
STEP_
UART_IO_
OA1#
OA2#
OB1#
OB2#
DRIVE_EN_
DIR_
STDBY
DRIVE_EN_
DIR_
STDBY
U_Control_Sheet
Control_Sheet.SchDoc
RXD
TXD
DTR
RTS
DRIVE_EN_
PWR_EN
CHG_
RXD
TXD
DTR
RTS
DRIVE_EN_
PWR_EN
CHG_
ADC_BAT
UART_IO_
STEP_
DIR_
TEMP_A
TEMP_B
STDBY
ADC_BAT
UART_IO_
STEP_
DIR_
TEMP_A
TEMP_B
STDBY
U_Connector Section
Connector Section.SchDoc
RXD
TXD
DTR
RTS
OA1#
OA2#
OB1#
OB2#
RXD
TXD
DTR
RTS
OA1#
OA2#
OB1#
OB2#
ADC_BAT
TS_
W_Pad1
TEMP_A
TEMP_B
ADC_BAT
TS_
W_Pad1
TEMP_A
TEMP_B

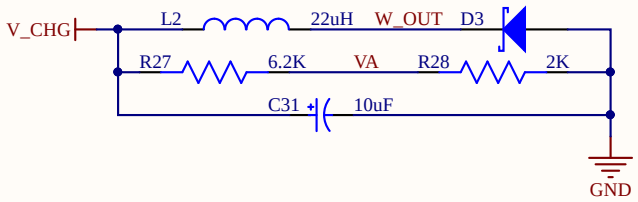

TS_
CHG_
PWR_EN
BOOT_ C1 22nF
FSW_ R1 150K
L1 10µH SW_
BAT C2 4.7uF
C3 4.7uF
8V C4 22uF
C5 22uF
V_CHG C6 4.7uF
VCC C8 1uF
GND
CHG_ RF1 10K
PWR_EN RF2 10K
FB_ R2 604K
R3 107K
ILIM_ R4 150K
3V3
8V
GND
C9 4.7nF
R5 18K COMP_
ISET_ R6 1.1K
GND
CF1 1uF
C10 0.1uF
BAT C11 4.7uF
3V3 C12 0.1uF
C13 1uF
GND
BAT
U1
9 VIN
11 SW
1 FSW
PWR_EN 7 EN
FB_ 3 FB
VOUT 6
VCC 2
COMP 4
BOOT 10
ILIM 8
GND 5
8V
VCC
COMP_
BOOT_
ILIM_
GND
TPS61089RNRT
V_CHG
U2
1 IN
4 PRETERM
ISET_ 2 ISET
GND 7 ISET2
TS_ 9 TS
6 NC
OUT 10
PG 5
CHG 8
VSS 3
GND 11
BAT
CHG_
GND
BQ24092DGQR
BAT
U3
7 IN
8 IN
3 N/C
BAT R7 6 EN
OUT 1
OUT 2
FB/ADJ 4
GND 5
GND 9
3V3
GND
NCP186AMX330TAG

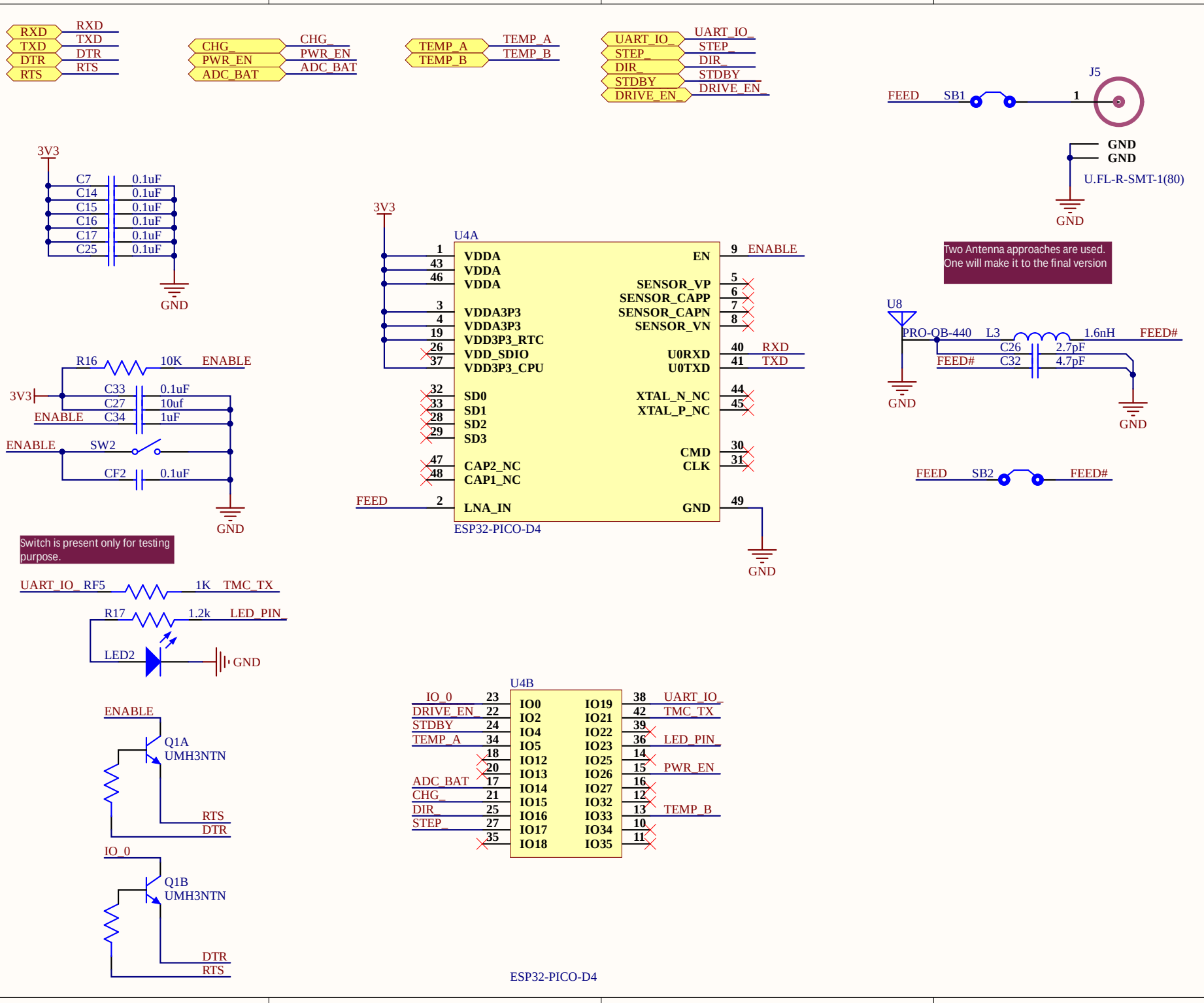

W_Pad1
W_Pad1

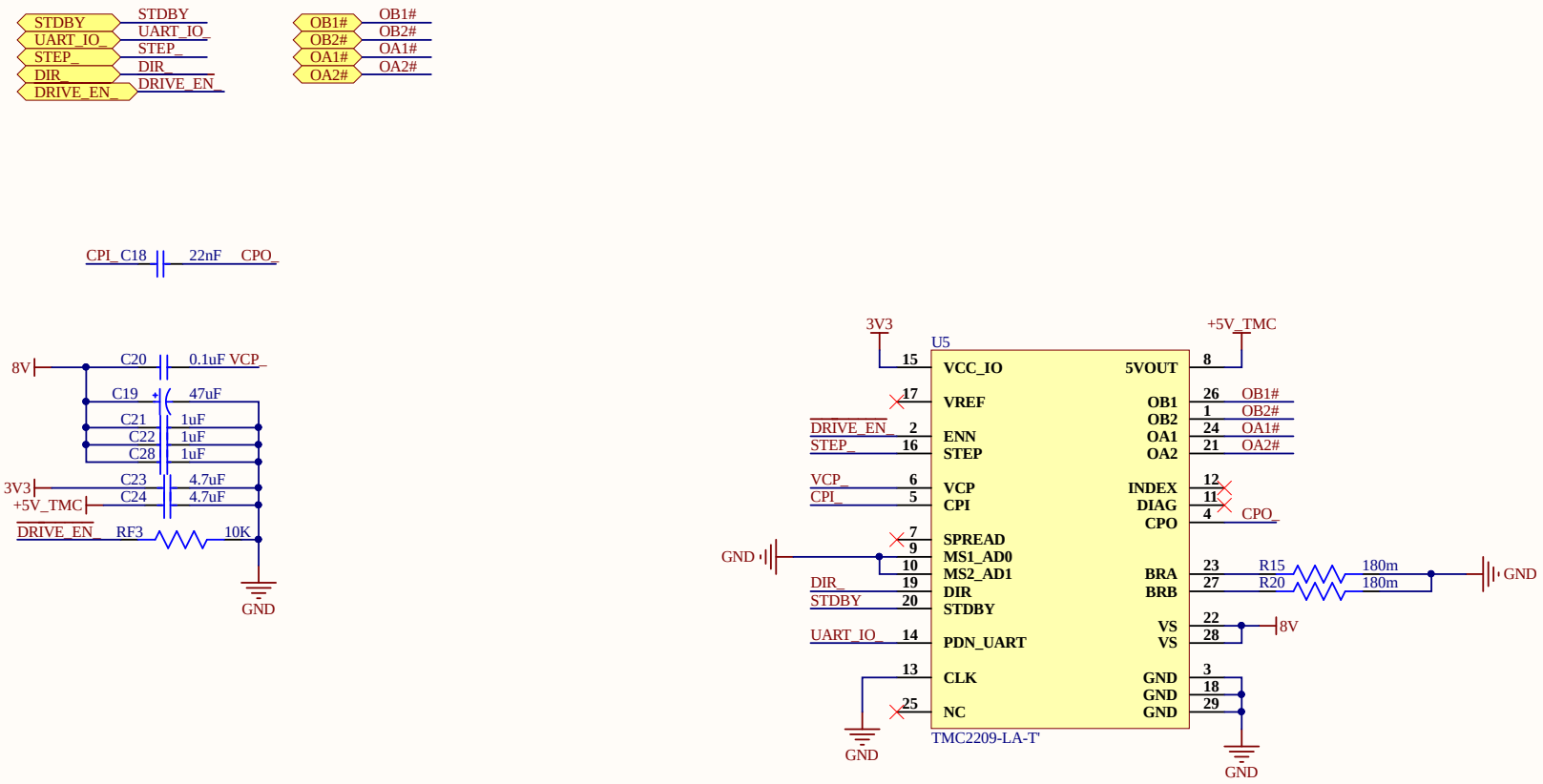

W_Pad1
C29
10000pF
D2
IN
C30
10uF
GND

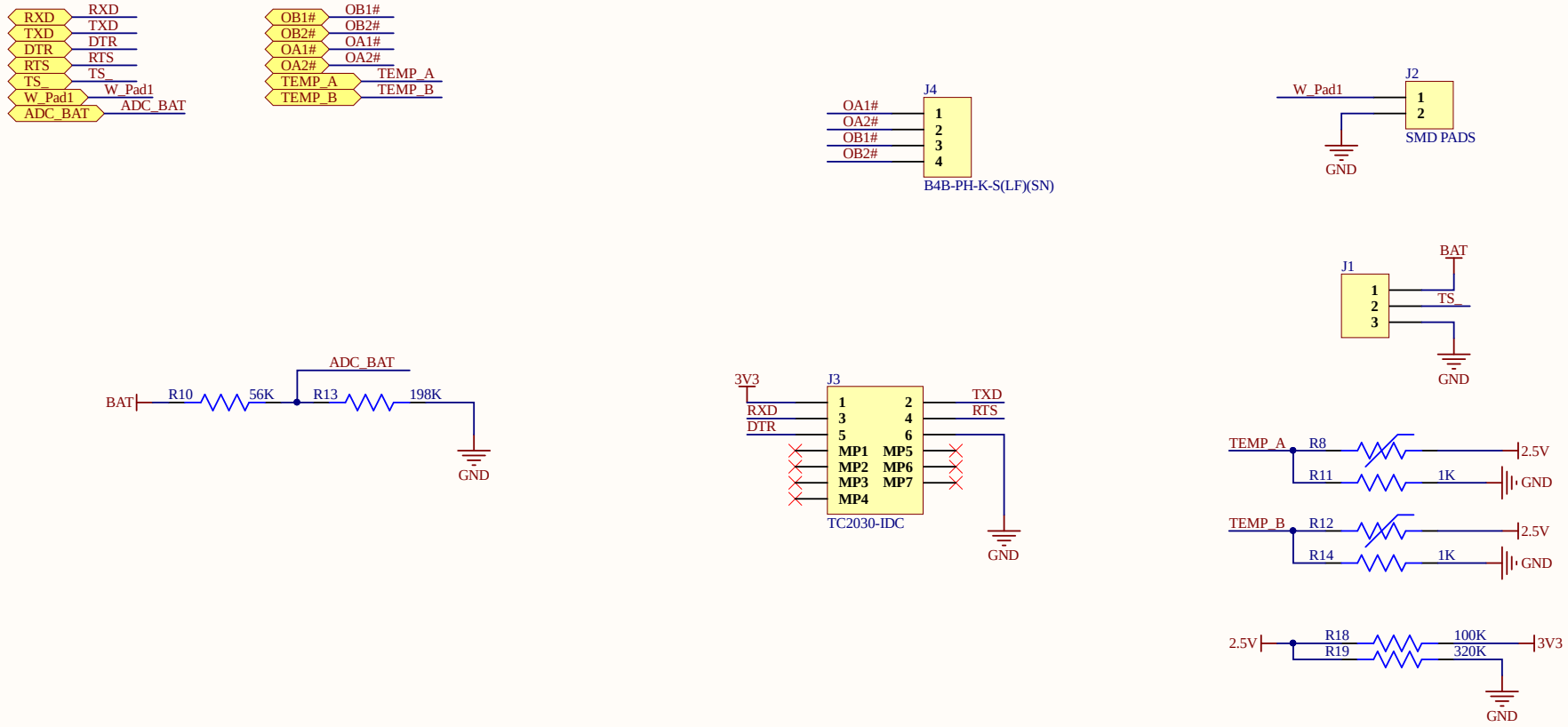

U7
IN
2
IN
OUT
3
W_OUT
VA
5
V A
O/F
7
IN#
1
NC
6
NC
8
NC
GND
4
XKT-510 T3168
GND


IN
R9
100K
IN#


V_CHG
L2
22uH
W_OUT
D3
R27
6.2K
VA
R28
2K
C31
10uF
GND


RXD
TXD
DTR
RTS
CHG_
PWR_EN
ADC_BAT
TEMP_A
TEMP_B
UART_IO_
STEP_
DIR_
STDBY
DRIVE_EN_
J5
FEED
SB1
GND
GND
U.FL-R-SMT-1(80)
3V3
C7 0.1uF
C14 0.1uF
C15 0.1uF
C16 0.1uF
C17 0.1uF
C25 0.1uF
U4A
VDDA
VDDA3P3
VDD3P3_RTC
VDD_SDIO
VDD3P3_CPU
SD0
SD1
SD2
SD3
CAP2_NC
CAP1_NC
LNA_IN
EN
SENSOR_VP
SENSOR_CAPP
SENSOR_CAPN
SENSOR_VN
U0RXD
U0TXD
XTAL_N_NC
XTAL_P_NC
CMD
CLK
ESP32-PICO-D4
Two Antenna approaches are used.
One will make it to the final version
U8
PRO-QB-440
L3 1.6nH
C26 2.7pF
C32 4.7pF
FEED#
R16 10K ENABLE
C33 0.1uF
C27 10uf
C34 1uF
SW2
CF2 0.1uF
Switch is present only for testing purpose.
UART_IO_ RF5 1K TMC_TX
R17 1.2k LED_PIN_
LED2
Q1A
UMH3NTN
Q1B
UMH3NTN
U4B
IO0
IO2
IO4
IO5
IO12
IO13
IO14
IO15
IO16
IO17
IO18
IO19
IO21
IO22
IO23
IO25
IO26
IO27
IO32
IO33
IO34
IO35
ESP32-PICO-D4


STDBY
UART_IO_
STEP_
DIR_
DRIVE_EN_
OB1#
OB2#
OA1#
OA2#
CPI_ C18 22nF CPO_
8V
C20 0.1uF VCP_
C19 47uF
C21 1uF
C22 1uF
C28 1uF
3V3
+5V_TMC
C23 4.7uF
C24 4.7uF
DRIVE_EN_ RF3 10K
GND
U5
3V3
+5V_TMC
15 VCC_IO
17 VREF
DRIVE_EN_ 2 ENN
STEP_ 16 STEP
VCP_ 6 VCP
CPI_ 5 CPI
7 SPREAD
9 MS1_AD0
10 MS2_AD1
DIR_ 19 DIR
STDBY 20 STDBY
UART_IO_ 14 PDN_UART
13 CLK
25 NC
5VOUT 8
OB1 26 OB1#
OB2 1 OB2#
OA1 24 OA1#
OA2 21 OA2#
INDEX 12
DIAG 11
CPO 4 CPO_
BRA 23 R15 180m
BRB 27 R20 180m
VS 22
VS 28
8V
GND 3
GND 18
GND 29
TMC2209-LA-T'
GND
GND
GND
GND


RXD
TXD
DTR
RTS
TS_
W_Pad1
ADC_BAT
OB1#
OB2#
OA1#
OA2#
TEMP_A
TEMP_B
J4
B4B-PH-K-S(LF)(SN)
J2
SMD PADS
GND
J1
BAT
3V3
J3
TC2030-IDC
MP1
MP2
MP3
MP4
MP5
MP6
MP7
R10
56K
R13
198K
R8
R11
1K
R12
R14
2.5V
R18
R19
100K
320K